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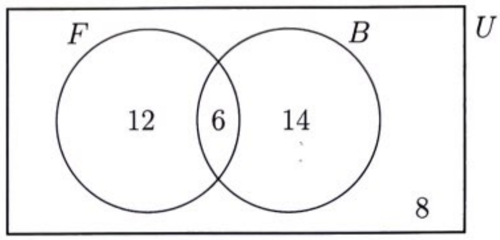
La Scuola d'Italia / Dr. Huson / IB Math: Probability Name: SOLUTIONS
5 March 2026

4.17 Quiz: Probability

1. In a group of 40 students, some play football (F) and some play basketball (B). Six students play both sports. This information is shown in the following Venn diagram.



   (a) Write down the number of students in the group who play football. [2]

   (b) One student from the group is chosen at random. Find the probability that

   (i) the student does not play football;

   (ii) the student plays football or basketball, but not both. [4]

a) 18

b) i) $\frac{14+8}{40} = \frac{22}{40}$

ii) $\frac{12+14}{40} = \frac{26}{40}$

2. Events A and B are mutually exclusive with $\mathrm{P}(A) = 0.55$ and $\mathrm{P}(B) = 0.15$.

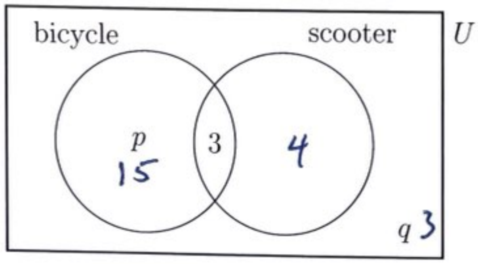
   (a) Write down $\mathrm{P}(A \cap B)$. [1]

   (b) Find $\mathrm{P}(A \cup B)$. [2]

a) 0

b) $0.55 + 0.15 = 0.7$

3. In a survey of 25 students, 18 own a bicycle and 7 own a scooter. Three students own both, as shown in the following Venn diagram. The values $\boldsymbol{p}$ and $\boldsymbol{q}$ represent numbers of students.



(a) Find the value of $\boldsymbol{p}$. [1]

(b) Find the value of $\boldsymbol{q}$. [1]

(c) A student is chosen at random. Find the probability that the student owns a scooter but not a bicycle. [2]

a) $p = 18 - 3 = 15$

b) $q = 25 - (15 + 3 + 4) = 3$

c) $\frac{4}{25}$

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4. The following table shows the sports and snack preferences of 20 students.

	Peanuts	Popcorn	Total
Mets fans	4	10	14
Yankees fans	2	4	6
Total	6	14	20

(a) Find the probability that a student chosen at random is a Yankees fan who likes popcorn. [2]

(b) Find the probability that a student is a Mets fan, given that they like peanuts. [2]

(c) Determine whether being a Mets fan and liking peanuts are independent events. Justify your answer. [2]

a) $\frac{4}{20}$

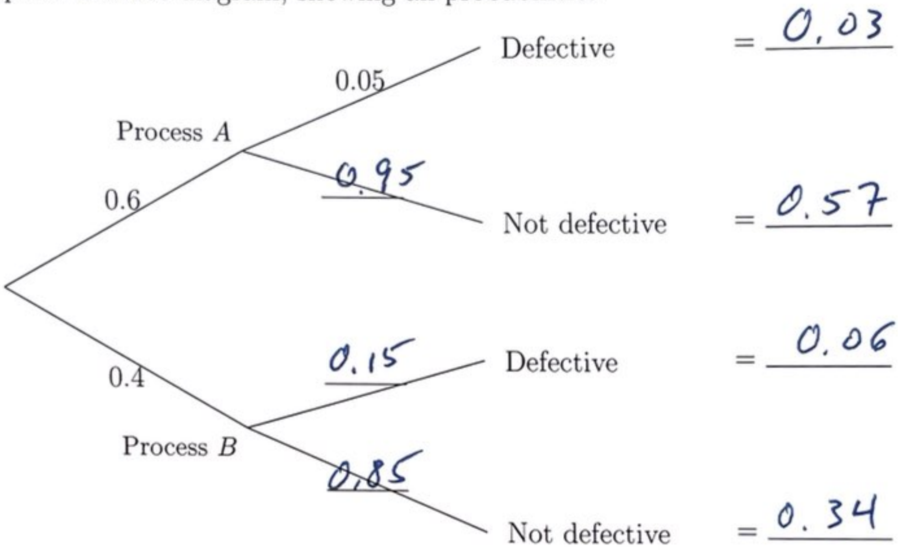
b) $\frac{4}{6}$

c) $\frac{4}{20} \neq \frac{14}{20} \cdot \frac{6}{20}$

NOT INDEPENDENT

5. A factory produces parts using two manufacturing processes. Process A produces 60% of all parts and process B produces the remaining 40%. The probability that a part from process A is defective is 0.05. The probability that a part from process B is defective is 0.15.

(a) Complete the tree diagram, showing all probabilities. [3]



(b) A part is selected at random. Find the probability that the part is

(i) defective and from process A; [2]

(ii) defective. [2]

(c) Given that a part is defective, find the probability that it came from process A. [3]

b) i) 0.03

ii) $0.03 + 0.06 = 0.09$

c) $\dfrac{0.03}{0.03+0.06} = \dfrac{1}{3}$

6. A drawer contains 8 black socks and 6 brown socks. Two socks are taken from the drawer at random, one after the other, without replacement.

   (a) Find the probability that both socks are black. [2]

   (b) Find the probability that the two socks are the same colour. [4]

a) $\frac{8}{14} \cdot \frac{7}{13} = \frac{28}{91}$

b) $P(\text{BROWN, BROWN}) = \frac{6}{14} \cdot \frac{5}{13} = \frac{15}{91}$

$\frac{28}{91} + \frac{15}{91} = \frac{43}{91}$

SOLUTIONS

4. [Maximum mark: 5]

Events A and B are such that $\mathrm{P}(A) = 0.4$, $\mathrm{P}(A|B) = 0.25$ and $\mathrm{P}(A \cup B) = 0.55$.

Find $\mathrm{P}(B)$.

$P(A) + P(B) - P(A \cap B) = P(A \cup B)$

$0.4 + P(B) - P(A \cap B) = 0.55$ A1

$P(A \cap B) = P(B) - 0.15$

$P(A|B) = \dfrac{P(A \cap B)}{P(B)} = 0.25$ A1

$P(B) - 0.15 = 0.25\,(P(B))$ A1

$0.75\,(P(B)) = 0.15$

$P(B) = 0.2$ A2

4.17 Probability - Early finishers Solutions

8.

a) i) $p = 12$

ii) $q = 100$

b) $P(A,A,A) = \frac{100}{160} \cdot \frac{99}{159} \cdot \frac{98}{158} = 0.241372\ldots \approx 0.241$

c) i) $P(A/m) = \frac{P(A \cap m)}{P(m)} = \frac{x}{48+x} = \frac{1}{3}$

$$3x = x + 48$$
$$x = 24$$

ii) $\frac{24}{160}$

d) NOT INDEPENDENT

$P(A)\,P(M) = \frac{100}{160} \cdot \frac{72}{160} \neq \frac{24}{160} = P(A \cap m)$

e) $P(\text{DARK CHOCOLATE}) = \frac{160-72}{160} = 0.55$

$\text{binomCDF}(10, 0.55, 5, 10) = 0.73843\ldots$
≈ 0.738